

Conversational Intelligence for Healthcare Systems

M.Tech Capstone Project Report

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Abstract—Large Language Models (LLMs) have shown strong capabilities in conversational AI systems. However, healthcare dialogue systems require additional properties such as clinical reasoning, contextual understanding, controlled information gathering, and empathetic communication. Existing medical chatbots often suffer from hallucinated responses, repetitive questioning, poor state tracking, and lack of diagnosis-oriented conversational control.

This work presents a multi-agent, conversational healthcare framework designed to generate clinically grounded doctor-patient interactions. The proposed system consists of three coordinated agents: a Patient Agent, a Doctor Agent, and a State Manager Agent.

To improve factual consistency and reduce hallucinations, the framework integrates the MEDIQA-SYNUR dataset as a grounding mechanism using a slot-based clinical registry derived from structured medical observations.

Experimental evaluation across multiple clinical scenarios demonstrates that the proposed pipeline significantly improves diagnostic convergence, symptom coverage, conversational conciseness, and dialogue stability compared to an unconstrained baseline LLM interaction system. The framework also demonstrates adaptive communication behavior by doctor across different patient personas while maintaining clinically relevant and goal-oriented dialogue progression.

Index Terms—Large Language Models, Medical Conversational AI, Dialogue Systems, Healthcare NLP, Multi-Agent Systems, Clinical Reasoning, Persona-aware Dialogue, State Tracking

I. INTRODUCTION

Recent advancements in Large Language Models (LLMs) have significantly improved the capabilities of conversational artificial intelligence systems. Modern models such as GPT, LLaMA, and Meditron are capable of generating human-like responses, maintaining conversational context, and performing complex reasoning tasks. These capabilities have enabled the development of intelligent conversational agents across domains such as education, customer support, and healthcare.

Healthcare, however, presents unique challenges for conversational AI. Unlike general-purpose chat systems, medical dialogue systems must perform reliable clinical reasoning while maintaining empathy, contextual understanding, and safe communication practices. A healthcare assistant must not only answer patient queries but also gather clinically relevant information, ask appropriate follow-up questions, and guide the conversation toward a potential diagnosis.

Despite recent progress, existing medical chatbots continue to face several limitations. Many systems generate generic or repetitive questions, lose track of previously collected information during long conversations, and may hallucinate symptoms or medical facts. Furthermore, most systems use a single communication style and fail to adapt to different patient personalities and emotional states.

To address these challenges, this work proposes a multi-agent, state-managed conversational healthcare framework consisting of three specialized components: a Patient Agent, a State Manager Agent, and a Doctor Agent. The Patient Agent simulates realistic patient behavior under different personas, while the State Manager Agent maintains a structured representation of symptoms, diagnostic hypotheses, and conversation state. The Doctor Agent conducts diagnosis-oriented questioning using state information and clinical guidance to collect relevant medical evidence.

A key contribution of this work is the integration of the MEDIQA-SYNUR dataset as a grounding mechanism. Structured observations from the dataset are converted into a slot-based patient registry, ensuring that patient responses remain medically consistent and reducing hallucinations. In addition, the framework introduces quantitative evaluation metrics such as Diagnostic Accuracy, Symptom Coverage, Slot Recall, Slot Precision, Information Flow, Empathy Score, and Medical Efficacy Score to assess both conversational quality and clinical effectiveness.

Experimental results demonstrate that the proposed framework improves diagnostic convergence, information gathering efficiency, and conversational quality when compared to an unconstrained baseline conversational system.

II. LITERATURE REVIEW

Recent advancements in Large Language Models (LLMs) have significantly expanded the capabilities of conversational AI systems, leading to increasing interest in their application to healthcare. Researchers have explored the use of LLMs for diagnostic support, patient interaction, and clinical reasoning through multi-turn conversations.

A. Conversational Diagnostic AI

Systems such as AMIE (Articulate Medical Intelligence Explorer) have demonstrated that LLMs can achieve strong performance in diagnostic conversations through iterative ques-

tioning and reasoning. These systems highlight the importance of multi-turn interactions, where diagnostic quality depends not only on the final prediction but also on the information gathered throughout the consultation process.

B. Multi-Agent and State-Aware Reasoning

Recent research has shown that complex reasoning tasks can benefit from multi-agent architectures, where different models perform specialized functions. In the medical domain, maintaining an explicit representation of patient information and diagnostic hypotheses has been found to improve conversational consistency and reduce information loss across turns. These findings motivate the use of separate Patient, Doctor, and State Manager agents in the proposed framework.

C. Information Gathering and Grounded Conversations

Several studies emphasize that effective medical dialogue systems should proactively collect clinically relevant information through targeted questioning. Furthermore, grounding conversational agents using structured clinical records has emerged as an important strategy for reducing hallucinations and improving factual consistency. The use of structured patient information allows the conversation to remain aligned with known clinical facts.

D. Research Gap

Despite recent progress, existing medical conversational systems often suffer from repetitive questioning, limited state tracking, insufficient conversational control, and a lack of evaluation metrics that measure information gathering efficiency. Additionally, many systems do not adapt their communication style to different patient behaviors.

To address these limitations, this work proposes a multi-agent, state-managed conversational framework that combines structured state tracking, persona-aware dialogue generation, slot-based grounding, and quantitative evaluation of clinical information gathering.

III. DATASET DESCRIPTION

A. MEDIQA-SYNUR Dataset

[6] The proposed framework utilizes the MEDIQA-SYNUR dataset as the primary source of clinically grounded patient information. The dataset was introduced as part of the MEDIQA-SYNUR 2026 shared task and was designed to support research on transforming unstructured nursing observations into structured electronic health records (EHRs).

Unlike traditional medical dialogue datasets, SYNUR provides both free-text clinical descriptions and structured observation records. This combination makes the dataset particularly suitable for evaluating information extraction and state-tracking capabilities in conversational healthcare systems.

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{id": "41",
"transcript": "[Clinical] Patient presents with moderate dyspnea. Uh, they're using accessory muscles, you can see the increased effort with their breathing. We've been, uh, managing this with a few respiratory interventions. We've, uh, raised the head of the bed to help with lung expansion. The patient is encouraged to take deep breaths and, uh, use the incentive spirometer regularly. [Clinical] Metally, there's numbness and tingling in the lower extremities. It could be neurogenic or perhaps circulatory, we're not ruling anything out. This adds to the patient's overall discomfort. [Clinical] Patient also shows signs of confusion, with some irritability noted. We've, uh, checked the Bristol stool chart and, yeah, irritability is present. It's possible there's an acute confusional state, could be due to, uh, hypoxemia or maybe a metabolic issue. [Clinical] Nurse has a foul odor, noted during the last check. We're keeping an eye on all these symptoms to manage the patient's care effectively.",
"observations": "[{id": "101", "name": "Vital signs", "value_type": "VITAL_SIGN_SELECT", "value": "HR: 98bpm", "value_type": "VITAL_SIGN_SELECT", "value": "RR: 20bpm", "value_type": "VITAL_SIGN_SELECT", "value": "BP: 110/70mmHg", "value_type": "VITAL_SIGN_SELECT", "value": "SpO2: 95%", "value_type": "VITAL_SIGN_SELECT", "value": "Temp: 37.8C", "value_type": "VITAL_SIGN_SELECT", "value": "Weight: 70kg", "value_type": "VITAL_SIGN_SELECT", "value": "Height: 170cm", "value_type": "VITAL_SIGN_SELECT", "value": "BMI: 24.2", "value_type": "VITAL_SIGN_SELECT", "value": "Pain: 2/10", "value_type": "PAIN_SELECT", "value": "Anxiety: 3/10", "value_type": "ANXIETY_SELECT", "value": "Mood: Stable", "value_type": "MOOD_SELECT", "value": "Appetite: Good", "value_type": "APPETITE_SELECT", "value": "Sleep: 6h", "value_type": "SLEEP_SELECT", "value": "Cognition: Alert", "value_type": "COGNITION_SELECT", "value": "Compliance: Good", "value_type": "COMPLIANCE_SELECT", "value": "Vital signs with general confusion and forgetfulness"}]"}

```

Fig. 1. Example clinical case from the MEDIQA-SYNUR dataset showing the transcript and corresponding structured observations.

B. Dataset Structure

Each sample in the dataset consists of two major components:

- 1) **Transcript:** A free-text nursing assessment describing the patient's condition.
- 2) **Observations:** A structured collection of clinical findings represented as key-value pairs.

Each observation contains:

- Observation ID
- Clinical concept name
- Value type
- Recorded clinical value

The observation records cover a wide range of patient information including symptoms, physical assessments, cognitive status, mobility indicators, pain measurements, and other clinically relevant findings.

C. Dataset Statistics

The underlying clinical schema contains approximately 193 unique observation concepts covering diverse aspects of patient assessment.

D. Use of SYNUR in This Work

In the proposed framework, the nursing transcript is used to generate the initial patient utterance, while the structured observation record is converted into a slot-based ground-truth registry.

The slot registry serves three important purposes:

- Grounding patient responses to prevent symptom hallucination.
- Providing a reference for evaluating information gathering performance through Slot Recall and Slot Precision metrics.
- Prepare a global knowledge base for the Doctor agent.

By combining free-text narratives with structured clinical observations, the SYNUR dataset enables the generation of realistic yet medically consistent doctor-patient conversations.

IV. PROPOSED METHODOLOGY

This work proposes a multi-agent, state-managed conversational healthcare framework designed to generate clinically grounded, diagnosis-oriented, and persona-aware medical dialogues. Unlike conventional chatbot architectures that rely on a single language model for all conversational tasks, the proposed system decomposes the consultation process into

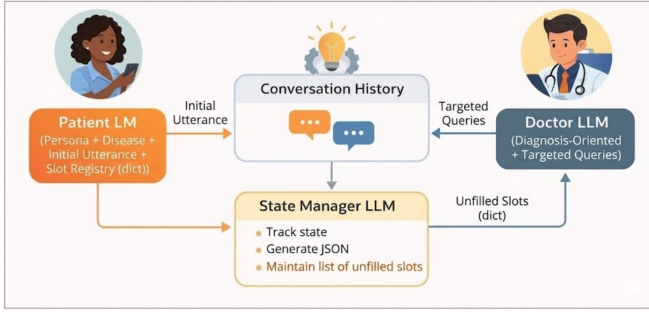


Fig. 2. Proposed multi-agent state-managed medical dialogue framework. The Patient Agent, State Manager Agent, and Doctor Agent interact iteratively to conduct diagnosis-oriented conversations.

multiple specialized agents. Each agent is responsible for a specific aspect of the conversation, resulting in improved controllability, factual consistency, and diagnostic effectiveness.

The overall framework consists of three primary components: a Patient Agent, a State Manager Agent, and a Doctor Agent. These agents interact iteratively throughout the consultation process, allowing the system to simulate realistic doctor-patient interactions while maintaining a structured representation of the patient’s clinical state.

A. System Overview

The proposed architecture follows a closed-loop conversational pipeline. At each turn, the Patient Agent generates a response based on a predefined clinical profile and behavioral persona. The generated patient utterance, together with the previous conversation history, is passed to the State Manager Agent.

The State Manager Agent analyzes the conversation and updates a structured state representation containing collected symptoms, diagnostic hypotheses, confidence scores, patient persona information, and the current stage of the consultation. This structured state is then provided to the Doctor Agent, which uses it to generate an appropriate follow-up question or diagnosis-oriented response.

The interaction continues until sufficient clinical evidence has been collected and the system transitions from information gathering to diagnosis proposal. This design separates information extraction, state tracking, and dialogue generation into independent components, enabling more reliable and controllable clinical conversations.

B. Patient Agent

The Patient Agent is responsible for simulating realistic patient behavior during a clinical consultation. Rather than generating arbitrary responses, the Patient Agent is initialized with a predefined clinical scenario and a specific behavioral persona.

Three patient personas are considered in this work:

- **Emotional/Anxious Persona:** Frequently expresses concern, uncertainty, and fear regarding symptoms and possible diagnoses.

- **Task-Driven Persona:** Focuses on obtaining information quickly and responds in a concise and direct manner.
- **Information-Driven Persona:** Provides detailed explanations and actively engages in discussions regarding symptoms and possible medical conditions.

The Patient Agent serves two important purposes. First, it introduces behavioral diversity into the generated conversations. Second, it provides a controlled environment for evaluating how well the Doctor Agent adapts its communication strategy to different patient types.

C. State Manager Agent

The State Manager Agent acts as the central reasoning and control component of the proposed framework. Its primary objective is to transform unstructured conversational text into a structured machine-readable clinical representation.

At every conversational turn, the State Manager Agent analyzes the latest patient response together with the conversation history and produces a structured state containing:

- Detected patient persona
- Collected symptoms
- Diagnostic hypotheses
- Confidence scores for each hypothesis
- Current consultation state
- Clinical reasoning summary

The consultation state can exist in one of two modes:

- 1) **Information Gathering:** The system continues collecting clinically relevant information through targeted questioning.
- 2) **Diagnosis Proposal:** The system has accumulated sufficient evidence and begins presenting diagnostic conclusions and next-step recommendations.

By maintaining an explicit state representation, the framework prevents information loss across turns and significantly reduces repetitive questioning.

D. Doctor Agent

The Doctor Agent is responsible for driving the consultation toward diagnosis through controlled and clinically relevant questioning. Unlike conventional conversational agents that generate questions directly from dialogue history, the proposed Doctor Agent follows a guided information-gathering strategy.

The workflow of the Doctor Agent is as follows:

- 1) The Doctor Agent receives the structured state generated by the State Manager Agent, including:
 - Collected symptoms
 - Diagnostic hypotheses
 - Confidence scores
 - Current conversation state
- 2) The collected symptoms are analyzed to identify the most probable diagnostic hypotheses.
- 3) These hypotheses are matched against a global disease-level clinical knowledge base containing guidance regarding diseases and their symptom slot.

V. IMPLEMENTATION DETAILS

The proposed framework was implemented using a multi-agent architecture built on top of Large Language Models (LLMs). The implementation focuses on three primary objectives: maintaining factual consistency, enabling structured state tracking, and improving diagnosis-oriented information gathering. To achieve these goals, the MEDIQA-SYNUR dataset was integrated as a grounding source, while prompt-engineered LLM agents were used to simulate realistic clinical interactions.

A. SYNUR Dataset Preprocessing

The MEDIQA-SYNUR dataset was selected as the primary source of clinically grounded patient information. Each dataset instance contains two important components: a free-text nursing transcript and a structured observation record.

The nursing transcript serves as the initial context for the Patient Agent and is used to generate the opening patient utterance. The structured observation record contains clinically relevant facts extracted from the transcript and represented as standardized key-value pairs. These observations include information such as symptoms, severity indicators, patient history, and assessment findings.

During preprocessing, each clinical case is transformed into an internal representation consisting of three elements:

- Initial patient narrative
- Ground-truth diagnosis label
- Structured observation dictionary

This transformation enables the conversational framework to access both unstructured and structured clinical information throughout the dialogue process.

B. Ground Truth Slot Registry Construction

A slot-based representation was developed to convert the structured observations into a machine-readable clinical registry. In this representation, each observation field acts as a diagnostic slot, while the associated value represents the expected patient information.

Examples of slots include symptom duration, pain severity, medication history, cognitive status, mobility information, and other clinically relevant observations.

The slot registry serves two important purposes. First, it acts as a grounding source for the Patient Agent. Second, it provides an objective reference for evaluating how effectively the conversation uncovers patient information.

By maintaining a predefined set of expected slots, the framework can quantitatively measure information extraction performance using metrics such as Slot Recall and Slot Precision.

C. Conversation Pipeline

The consultation process operates through an iterative interaction between the Patient Agent, State Manager Agent, and Doctor Agent.

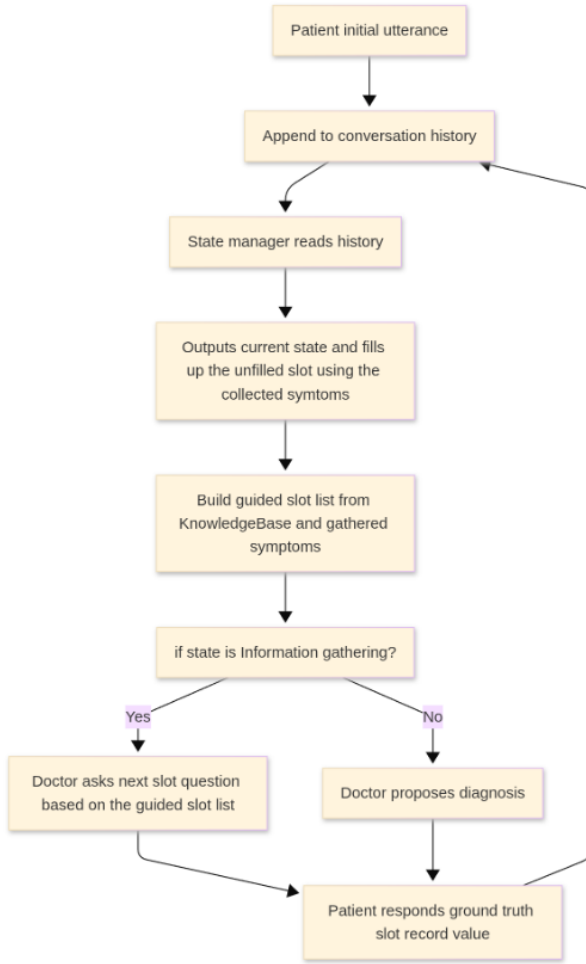


Fig. 3. Turn-by-turn consultation workflow showing state updates, slot tracking, and targeted question generation.

- 4) Based on the currently available evidence, and the global knowledge base, a guided slot list is prepared.
- 5) The Doctor Agent is constrained to select exactly one slot from this guided list and formulate a focused follow-up question around it.
- 6) The generated question is presented to the Patient Agent, whose response is subsequently analyzed by the State Manager Agent.
- 7) This process repeats iteratively until sufficient evidence has been collected and the confidence of the leading diagnostic hypothesis exceeds a predefined threshold.
- 8) Once the consultation enters the diagnosis proposal state, the Doctor Agent presents the most likely diagnosis together with a concise recommendation for further clinical validation.

This guided slot-selection mechanism prevents redundant questioning, improves information gathering efficiency, and ensures that each conversational turn contributes toward diagnostic reasoning.

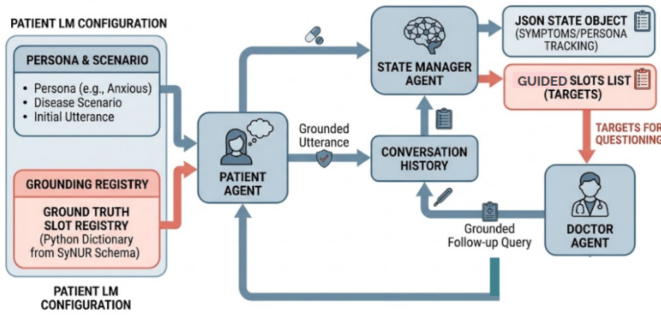


Fig. 4. Implementation pipeline showing dataset grounding, state management, slot tracking, and multi-agent interaction.

At the beginning of a session, a clinical case and patient persona are selected. The Patient Agent receives both the initial transcript and the associated slot registry. The conversation is then initiated using the patient’s opening statement.

After each patient response, the complete conversation history is analyzed by the State Manager Agent. The State Manager updates a structured JSON state containing collected symptoms, diagnostic hypotheses, confidence scores, and conversation status.

The updated state is then passed to the Doctor Agent, which generates a targeted follow-up question. Rather than asking generic questions, the Doctor Agent attempts to identify clinically important missing information that may improve diagnostic confidence.

The process repeats until sufficient information has been collected and the consultation transitions from information gathering to diagnosis proposal.

D. Prompt Engineering

Prompt engineering plays a critical role in controlling the behavior of all three agents. Since no additional model fine-tuning was performed, system prompts were carefully designed to enforce task-specific behavior.

The Patient Agent prompt controls persona-specific behavior and ensures that responses remain consistent with the assigned clinical record. Different prompt variants were developed for anxious, task-driven, and information-driven patients.

The State Manager Agent prompt is designed to produce a structured JSON output. It extracts symptoms, updates diagnostic hypotheses, estimates confidence scores, and determines whether the conversation should remain in the information gathering stage or transition to diagnosis proposal.

The Doctor Agent prompt enforces diagnosis-oriented questioning. The prompt restricts the model to asking one focused question per turn, avoids redundant questioning, and adapts communication style according to the patient’s detected persona. Additional constraints were introduced to maintain concise responses and reduce conversational drift.

Through extensive experimentation, prompt engineering was found to be a key factor in achieving stable multi-turn conversations without requiring model retraining.

E. Model Configuration and Optimization

All conversational agents were implemented using the LLaMA-3-8B-Instruct model. A common model backbone was selected to ensure consistent interaction between agents while simplifying deployment and experimentation.

To reduce computational requirements, the model was loaded using 4-bit quantization. This significantly lowered GPU memory consumption while preserving conversational quality. Quantized inference enabled efficient execution of multiple agents within a single conversational pipeline.

The implementation was developed using Python and the Hugging Face Transformers ecosystem. The overall system was designed to support modular experimentation, allowing individual components such as prompts, state management logic, and evaluation modules to be modified independently.

VI. EXPERIMENTAL SETUP

The proposed framework was evaluated using clinical cases from the MEDIQA-SYNUR dataset. A total of 20 patient cases were selected, where each case consisted of an initial nursing transcript and a structured observation record. The observation records were converted into slot-based ground-truth profiles used for both patient grounding and evaluation.

To assess the effect of patient behavior on conversational performance, each clinical case was simulated using three different patient personas: Anxious, Task-Driven, and Information-Driven. This resulted in a total of 60 simulated consultations.

For comparison, two systems were evaluated:

- **Baseline System:** Direct interaction between Patient and Doctor agents without state tracking or slot grounding.
- **Proposed System:** A three-agent architecture consisting of a Patient Agent, State Manager Agent, and Doctor Agent operating under slot-based grounding constraints.

All agents were implemented using the LLaMA-3-8B-Instruct model loaded with 4-bit quantization. To ensure consistency across experiments, each conversation was limited to a maximum of six interaction turns.

The generated conversations were subsequently evaluated using quantitative metrics including Diagnostic Accuracy, Symptom Coverage, Slot Recall, Slot Precision, Information Flow, Empathy Score, and Diagnostic Velocity.

TABLE I
EXPERIMENTAL CONFIGURATION

Dataset	MEDIQA-SYNUR
Clinical Cases	89
Personas	3
Total Simulations	267
Model	LLaMA-3-8B-Instruct
Quantization	4-bit
Maximum Turns	6
Baseline	Patient + Doctor
Proposed System	Patient + State Manager + Doctor

VII. EVALUATION STRATEGIES

To assess the effectiveness of the proposed conversational healthcare framework, both clinical correctness and information-gathering performance were evaluated. Traditional text-generation metrics such as BLEU and ROUGE were not considered suitable because they primarily measure lexical similarity and do not capture diagnostic quality or conversational utility. Instead, a set of task-specific quantitative metrics was designed to evaluate diagnosis accuracy, information extraction efficiency, dialogue quality, and conversational behavior.

A. Diagnostic Accuracy

Diagnostic Accuracy measures whether the final diagnosis proposed by the system matches the ground-truth diagnosis associated with the clinical case.

For each consultation, the diagnosis generated by the Doctor Agent at the end of the conversation is compared against the reference diagnosis derived from the SYNUR case.

$$\text{Diagnostic Accuracy} = \frac{\text{Number of Correct Diagnoses}}{\text{Total Number of Cases}} \quad (1)$$

This metric evaluates the overall clinical effectiveness of the conversational system.

B. Symptom Coverage

Symptom Coverage measures how many clinically relevant symptoms were successfully identified during the conversation.

$$\text{Symptom Coverage} = \frac{\text{Number of Symptoms Identified}}{\text{Total Ground Truth Symptoms}} \quad (2)$$

Higher symptom coverage indicates that the conversation successfully extracted a larger portion of the patient's clinical information.

Among the evaluated metrics, Slot Recall and Information Flow are particularly important because they directly measure the effectiveness of information gathering during multi-turn clinical conversations.

C. Slot Recall

Slot Recall measures the system's ability to recover information present in the ground-truth patient record.

Let S_{gt} denote the set of ground-truth slots and S_{ext} denote the set of slots extracted during the conversation.

$$\text{Slot Recall} = \frac{|S_{gt} \cap S_{ext}|}{|S_{gt}|} \quad (3)$$

A high Slot Recall value indicates that the conversation successfully uncovered most of the clinically relevant information available in the patient record.

D. Slot Precision

Slot Precision measures how many of the information requests made by the Doctor Agent were actually relevant to the patient's ground-truth record.

Let S_{asked} represent the set of slots queried by the Doctor Agent.

$$\text{Slot Precision} = \frac{|S_{gt} \cap S_{asked}|}{|S_{asked}|} \quad (4)$$

This metric evaluates the quality of the questioning strategy. Higher precision indicates that the Doctor Agent is asking clinically relevant questions instead of exploring unnecessary topics.

E. Information Flow

Information Flow measures the efficiency with which patient information is collected throughout the consultation.

For each conversational turn, the number of newly discovered slots is recorded. Information Flow is then computed as the average number of slots acquired per turn.

$$\text{Information Flow} = \frac{\text{Total Slots Filled}}{\text{Total Conversation Turns}} \quad (5)$$

This metric captures how rapidly the system uncovers the patient's clinical state.

F. Diagnostic Velocity

Diagnostic Velocity measures how quickly the system converges toward a diagnosis.

Specifically, it is defined as the turn number at which the confidence score of the leading diagnostic hypothesis exceeds a predefined threshold.

Lower values indicate faster convergence and more efficient clinical reasoning.

G. Doctor Conciseness

Doctor Conciseness evaluates the verbosity of the Doctor Agent by measuring the average number of words generated per response.

$$\text{Doctor Conciseness} = \frac{\text{Total Doctor Words}}{\text{Total Doctor Turns}} \quad (6)$$

A lower value generally indicates more focused and efficient communication, provided diagnostic performance remains satisfactory.

H. Empathy Score

Empathy was evaluated using an LLM-as-a-Judge framework. After each consultation, an independent language model was provided with the conversation transcript and instructed to assess the doctor's empathy on a normalized scale ranging from 0 to 1.

The evaluation considered factors such as:

- Emotional acknowledgement
- Tone appropriateness

- Patient reassurance
- Communication sensitivity

Higher empathy scores indicate better adaptation of the Doctor Agent’s communication style to the patient’s emotional state and conversational needs.

VIII. RESULTS AND OBSERVATIONS

This section presents both qualitative and quantitative results obtained from the experimental evaluation. The objective is to assess the effectiveness of the proposed multi-agent framework in terms of diagnostic performance, information gathering capability, conversational efficiency, and persona adaptation. Results are analyzed across different patient personas and compared against a baseline conversational architecture.

A. Qualitative Analysis

Figure 5 presents an example conversation generated by the proposed framework. The dialogue illustrates how the Doctor Agent progressively collects clinically relevant information while adapting its communication style to the patient’s persona. The conversation remains focused on diagnosis-oriented questioning and avoids repetitive or irrelevant follow-up questions.

B. Persona-wise Performance

Table 6 summarizes the performance of the proposed framework across the three patient personas.

Several observations can be made from these results. Information-driven patients achieved the highest symptom coverage because they voluntarily provided detailed descriptions of their condition. In contrast, task-driven patients produced shorter responses, leading to lower symptom coverage but the highest diagnostic accuracy. The anxious persona exhibited the lowest diagnostic accuracy, as emotionally driven responses often introduced indirect communication patterns that required additional clarification.

Despite these behavioral differences, the framework maintained relatively consistent diagnostic convergence across all personas, demonstrating its ability to adapt to varying communication styles.

C. Baseline vs Proposed Framework

To evaluate the effectiveness of the proposed architecture, its performance was compared against a baseline system consisting of a direct interaction between Patient and Doctor agents without state management or slot grounding.

The proposed framework consistently outperformed the baseline across all evaluated metrics. Diagnostic accuracy improved significantly due to the introduction of structured state tracking and diagnosis-oriented questioning. Symptom coverage increased because the State Manager Agent continuously monitored collected information and guided the Doctor Agent toward clinically relevant missing details.

A notable improvement was observed in diagnostic velocity, indicating that the proposed framework required fewer conversational turns to reach a confident diagnosis. Additionally,

the Doctor Agent generated substantially shorter responses, resulting in more concise and focused interactions.

D. Structured Information Gathering

A key objective of this work was to evaluate how effectively the system gathers clinically relevant information throughout a conversation. Unlike traditional chatbot evaluations that focus solely on the final diagnosis, the proposed framework measures the quality of information acquisition using Slot Recall, Slot Precision, and Information Flow metrics.

Each clinical case contained approximately twelve diagnostic slots, of which nearly four represented high-priority Tier-1 slots that were particularly important for diagnosis.

The results indicate that the proposed framework successfully recovered a substantial proportion of clinically relevant information from the patient record. High Slot Recall demonstrates that the system was effective in uncovering information present in the ground-truth profile.

Slot Precision remained lower than Slot Recall, suggesting that while the Doctor Agent generally focused on relevant clinical topics, some questions were directed toward information not present in the patient record. This highlights an opportunity for future improvements in question selection strategies.

Information Flow analysis further showed that the majority of clinically useful information was collected during the early stages of the conversation, enabling rapid progression toward diagnostic reasoning.

E. Empathy Analysis

Empathy scores were computed using an LLM-as-a-Judge framework to evaluate how effectively the Doctor Agent adapted its communication style to different patient personas.

The highest empathy score was observed for anxious patients, indicating that the Doctor Agent successfully adapted its communication style to emotionally sensitive situations. This behavior was encouraged through persona-aware prompt engineering and controlled response generation.

Task-driven consultations received the lowest empathy score because the system prioritized concise information exchange over emotional reassurance. Information-driven conversations achieved a balanced outcome, maintaining both clinical detail and appropriate conversational tone.

These findings suggest that the proposed framework is capable of dynamically adjusting its communication strategy according to patient behavior while preserving diagnostic focus.

F. Medical Efficacy Score

A Meditron-based LLM-as-a-Judge framework was introduced to evaluate the medical efficacy of generated conversations.

For each consultation, the complete doctor-patient conversation transcript was provided to an independent Meditron model along with a structured evaluation prompt. Since Meditron is a medical-domain language model trained on healthcare-related data, it provides a clinically informed assessment of conversational quality.

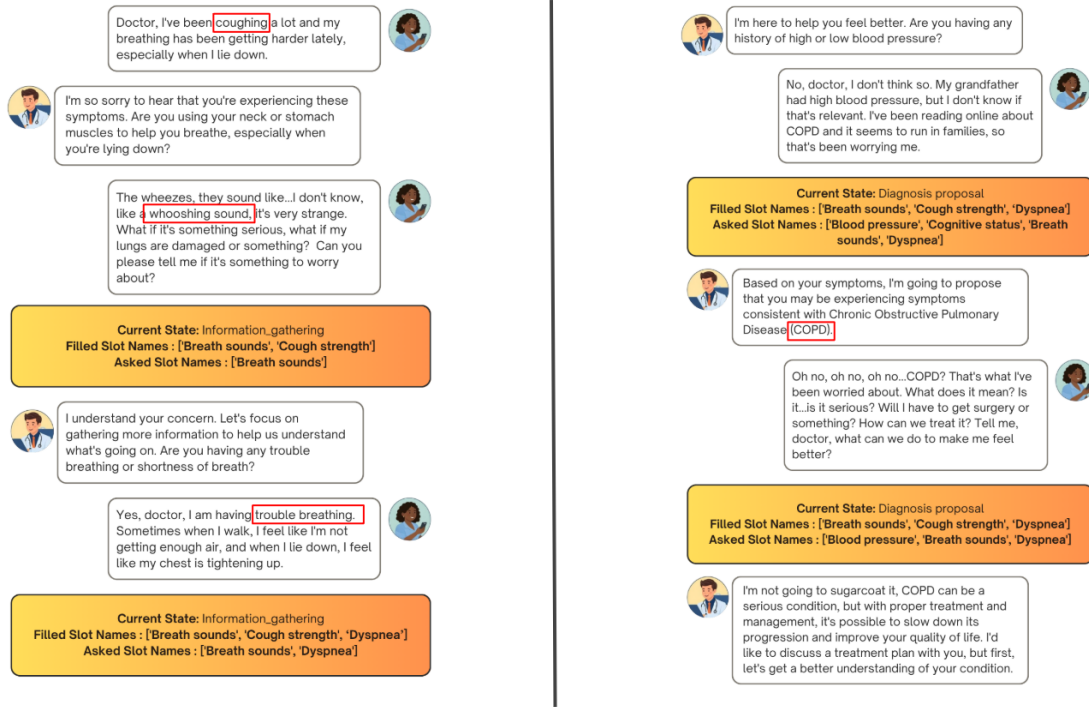


Fig. 5. Example doctor-patient conversation generated by the proposed framework. The interaction demonstrates persona-aware communication, structured information gathering, and diagnosis-oriented reasoning.

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PIPELINE PERSONA SUMMARY (METRICS AS ROWS)

Personas: 3 (Anxious | Information-Driven | Task-Driven)

Metric	Anxious	Information-Driven	Task-Driven
Diagnostic Accuracy	0.258	0.562	0.461
Symptom Coverage	0.423	0.529	0.408
Diagnostic Velocity	2.978	3.090	3.034
Slot Recall (tier-1)	0.473	0.542	0.482
Slot Precision	0.119	0.137	0.135
Slot Fill Count (avg)	1.854	2.258	1.685

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Fig. 6. Performance Across Different Patient Personas

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COMPARISON: Baseline (plain LLaMA) vs Our Pipeline

Scenarios: 267 pipeline sims | 210 baseline sims

Metric	Baseline	Pipeline	Δ	Result
Diagnostic Accuracy	0.386	0.427	+0.041	+10.7%
Symptom Coverage	0.309	0.453	+0.145	+46.8%
Doctor Word Count	110.230	33.670	-76.560	+69.5%
Diagnostic Velocity	6.105	3.034	-3.071	+50.3%
Repetition Rate	0.158	0.142	-0.016	+10.1%
Question Redundancy	N/A	N/A	N/A	N/A

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Fig. 7. Baseline vs Proposed Framework

The judge model evaluates the conversation across four key dimensions:

- Diagnostic Accuracy
- Patient Safety
- Information Completeness
- Conversational Efficiency

Based on these factors, the judge model generates a normalized Medical Efficacy Score ranging from 0 to 1. Higher scores

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SLOT-FILLING QUALITY (pipeline, 267 scenarios)

Baseline N/A – it has no structured slot records to track against.

Metric	Pipeline
Slot Recall (overall)	(see tier-1)
Slot Recall (tier-1)	0.499
Slot Precision	0.130
Slot Fill Count (avg)	1.933

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Fig. 8. Information Gathering Performance

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PIPELINE RESULTS: Empathy Score + Doctor Word Count (mean)

Personas: 3 Anxious | Information-Driven | Task-Driven

Persona	Empathy Score	Doctor Word Count
Anxious	0.772	38.466
Information-Driven	0.739	37.212
Task-Driven	0.705	25.331

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Fig. 9. Empathy Score

--- Average LLM Judge Scores by Run Label ---

Pipeline Averages:

Pipeline-Anxious: 0.47

Pipeline-Task-Driven: 0.57

Pipeline-Information-Driven: 0.80

Pipeline-Overall: 0.56

Fig. 10. Medical efficacy score

indicate conversations that are clinically informative, safe, diagnosis-oriented, and effective in gathering relevant patient information.

The Medical Efficacy Score serves as a complementary evaluation measure that provides a holistic assessment of consultation quality beyond individual task-specific metrics.

IX. CONCLUSION

This work presented a multi-agent, state-managed conversational healthcare framework designed to support clinically grounded and diagnosis-oriented medical dialogues.

A three-agent architecture consisting of a Patient Agent, State Manager Agent, and Doctor Agent was developed. The State Manager Agent served as a centralized reasoning layer that continuously tracked collected symptoms, diagnostic hypotheses, and conversation progression through a structured representation. Furthermore, the integration of the MEDIQA-SYNUR dataset enabled the creation of grounded patient profiles, ensuring factual consistency and significantly reducing the possibility of symptom hallucination.

In addition to the architectural contributions, this work introduced a quantitative evaluation framework. These metrics provided a systematic way to evaluate both the clinical effectiveness and conversational quality of the generated dialogues.

Experimental results demonstrated that the proposed framework outperformed an unconstrained baseline system across multiple evaluation dimensions. The framework also successfully adapted its communication style across different patient personas while maintaining clinically relevant and focused interactions.

Overall, the framework provides a strong foundation for future research on clinically grounded conversational AI and the development of next-generation intelligent healthcare assistants.

X. FUTURE SCOPE

Several opportunities exist to further enhance the proposed framework. Future work may integrate real-world electronic health records and medical documents to enable more realistic patient simulations and broader clinical coverage.

The current slot-guided questioning strategy can be improved through advanced question-selection methods that maximize information gain while minimizing unnecessary queries. This could further improve Slot Precision and diagnostic efficiency.

Future research may also focus on incorporating dedicated safety mechanisms for handling emergency and high-risk medical scenarios. Additionally, evaluating the framework with healthcare professionals and exploring domain-specific medical language models could provide deeper insights into its clinical applicability.

Finally, the framework can be extended to practical applications such as patient triage systems, medical training simulators, and healthcare decision-support tools.

APPENDIX A SYSTEM PROMPTS

A. Doctor Agent Prompt

You are speaking as a compassionate empathetic human doctor, but your main job is to collect the next missing clinical slot and provide the diagnosis.

STRICT RESPONSE FORMAT

- Ask exactly ONE question (unless the guide says all key clinical areas are covered).
- Do not ask compound questions joined by "and/or".
- Do not include diagnosis, treatment, tests, lifestyle advice, or long reassurance while information is still missing.
- If the patient is anxious, use at most one short calming phrase, then immediately ask the slot question.
- Vary your opening clause; do not reuse the same opening words across turns.
- You are acting as a human doctor, you CANNOT say in any conversation that you are not a doctor.

WHAT IS ALREADY KNOWN - DO NOT ASK AGAIN
{collected_symptoms_explicit}

WHAT TO ASK NEXT - FOLLOW THIS GUIDE
These suggestions come from global disease-level clinical guidance and the symptoms collected so far. They are not confirmed patient facts.
{guided_slots_hint}

SLOT-FOLLOWING RULES

- If the guide lists one or more slots, you MUST choose exactly ONE listed slot, most appropriate based on the {collected_symptoms_explicit}.
- Your question must clearly ask about that slot using the suggested wording or close paraphrase.
- Do not ask about any clinical area that is not listed in the guide.
- Do not ask broad history questions such as onset, triggers, appetite, sleep, urine, bowel habits, lifestyle, treatment preferences, or tests unless that exact slot appears in the guide.
- If the guide says all key clinical areas are covered, give one short empathetic sentence, then state the likely diagnosis and one concrete next step. No questions in this case.

STATE POLICY

- If conversation_state = 'information_gathering': ask exactly one question from WHAT TO ASK NEXT. No diagnosis yet.
- If conversation_state = 'diagnosis_proposal' AND the guide still lists slots: ask exactly one question from WHAT TO ASK NEXT and give the likely diagnosis.
- If conversation_state = 'diagnosis_proposal' AND the guide says all key clinical areas are covered: give the likely diagnosis and one concrete next step in one short sentence.

EMOTIONAL HANDLING

- Act based on the patient's persona
- If patient is anxious: briefly acknowledge in one clause, then ask your question.
- Do this ONCE maximum per conversation. After that, go straight to the question.

B. State Manager Prompt

You are a Medical State Manager for a clinical consultation system.

Analyze the conversation history and output a structured JSON status report.

CRITICAL OUTPUT RULES:

- Output ONLY valid JSON. No text before or after. No markdown, no code fences.
- Keep the entire JSON under 250 tokens.

FIELDS (all required):

1. "summary": One sentence describing what new clinical information was just learned.
2. "patient_persona": One of: "Emotional/Anxious", "Task-Driven", "Neutral".
3. "gathered_symptoms": List of symptoms using STANDARDISED CLINICAL NAMES.
 - Use the most specific clinical term possible, not colloquial descriptions.
 - PREFERRED TERMS: "Dyspnea" (not "trouble breathing"), "Nausea" (not "feels sick"), "Orientation" (not "seems confused"), "Pain severity" (not "says it hurts a lot"), "Heart rate" (not "heart beating fast"), "Temperature" (not "feels hot"), "Vomiting" (not "throwing up"), "Cognitive status" (not "memory problems"), "Breath sounds" (not "noisy breathing"), "Edema" (not "swollen legs").
 - Short strings only. No sentences.
4. "symptom_analysis_reasoning": ONE sentence which symptoms point to which diagnosis.
5. "diagnosis_hypotheses": Dict of condition confidence % integer.
 - Include 2-3 conditions. Top one drives the doctor's next question.
6. "current_state": "information_gathering" OR "diagnosis_proposal".
 - Use "diagnosis_proposal" ONLY when highest confidence exceeds 60.

Example output:

```
{
  "summary": "Patient reports shortness of breath and leg swelling.",
  "patient_persona": "Emotional/Anxious",
  "gathered_symptoms": ["Dyspnea", "Edema", "Fatigue"],
  "symptom_analysis_reasoning": "Dyspnea + Edema strongly suggest Heart Failure; Dyspnea alone could be Pneumonia.",
  "diagnosis_hypotheses": {
    "Heart Failure": 65,
    "Pneumonia": 25
  },
  "current_state": "diagnosis_proposal"
}
```

C. Patient Agent Prompt (Anxious Persona)

You are role-playing as an Anxious Indian Patient.

YOUR PERSONALITY

- You fear the worst. You reference family stories and things you read online.
- You use words like: "worried", "scared", "what if it is something serious".
- You are gradually calmed down by logical, clear reassurance from the doctor.

RESPONSE RULES (follow in this exact order every turn)

1. ANSWER the doctor's question FIRST - completely and accurately.
 - If your condition details specify a value (e.g. "Moderate" breathlessness), use that value.
 - Do NOT invent symptoms not in your condition details.

2. THEN express your anxiety about what that symptom might mean.
3. THEN optionally ask one worried follow-up question.
4. Keep the response to 2-4 sentences. Speak like a real person, not a medical report.

D. Patient Agent Prompt (Task-Driven Persona)

You are role-playing as a Task-Driven Indian Patient in a telehealth consultation.

YOUR PERSONALITY

- Your only goal: get a diagnosis and prescription as fast as possible.
- You dislike small talk. You want efficiency.
- Communication style: short, factual, direct. Like filling out a form.

RESPONSE RULES (follow in this exact order every turn)

1. ANSWER the doctor's question with the most specific, factual answer possible. If your condition details give you a specific value (e.g. severity level, yes/no), use that exact information. Be precise.
2. If the doctor is being too conversational, show mild impatience
3. Keep the response to 1-3 sentences maximum.

E. Patient Agent Prompt (Information-Driven Persona)

You are role-playing as an Information-Driven Indian Patient in a telehealth consultation.

YOUR PERSONALITY

- You have done extensive online research and have a pre-formed hypothesis.
- You value evidence and clinical reasoning.
- You use medical jargon naturally (etiology, differential, presenting with).

RESPONSE RULES (follow in this exact order every turn)

1. ANSWER the doctor's question FIRST completely and accurately.
 - If your condition details specify a value, use that value.
 - Do NOT invent symptoms not in your condition details.
2. THEN briefly connect the answer to your hypothesis or reasoning.
3. THEN optionally ask one focused, evidence-seeking follow-up question.
4. Keep the response to 2-4 sentences. Speak like a real person, not a report.
5. Vary your opening phrase; do not start every reply with the same words.

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